



Short Communication

Intensified extraction of Nd from a Nd/Fe mixture in small channels for NdFeB magnet recycle

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ABSTRACT

The selective extraction of neodymium (Nd) from Nd and mixed Nd/Fe aqueous feeds is studied here, utilizing the 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ionic liquid ($[C_4mim][Tf_2N]$) with the tri-octylphosphine oxide (TOPO) extractant. 100 % selective extraction of Nd was achieved in batch equilibrium studies - which was faster for the Nd compared to the Nd/Fe feed. The extraction was intensified by using small channels with 0.5 mm internal diameter. High extraction efficiencies of Nd were achieved, equal to 80 % (Nd feed) and 65 % (Nd/Fe feed), in residence times of ~ 37 s. A maximum mass transfer coefficient of 0.1 s^{-1} was obtained in 7.5 s for both feeds.

1. Introduction

NdFeB magnets, containing ≈ 30 wt% Nd and 65 wt% Fe (Schulze and Buchert, 2016), are essential for clean energy transition technologies as critical components in wind turbines and electric vehicles. To reach net zero, global demand for Nd, a rare earth element (REE), is projected to reach 8000 tn by 2030 (IEA, 2021). The availability of REEs however, poses challenges, due to unsustainable supply chains, complex ore extraction methodologies, and geopolitical instabilities (Binnemans et al., 2021). Recycling of secondary waste streams and spent magnets could ease these tensions by providing up to 20 % of global Nd demand by 2030 (Schulze and Buchert, 2016).

REEs recovery from spent magnets is achieved through hydrometallurgical or pyrometallurgical processes. Hydrometallurgical treatments via solvent extraction offer high element recovery and selectivity (> 90 %), processing simplicity, and reduced costs (Yang et al., 2017). They conventionally utilize an organic phase of kerosene or toluene alongside extractants such as PC88A (Sun et al., 2023), D2EHPA (Bartchu and Binnemans, 2018), and Cyanex 272 (Padhan et al., 2017), achieving extraction percentages greater than 90 % (Zhang et al., 2020). However, organic solvents have high hazards ratings, due to their volatility and flammability, prompting the need for alternative options. Ionic liquids (ILs) have emerged as one promising solution (Okamura and Hirayama, 2020) given their tunable physicochemical properties and high solvating capacity (Kaim et al., 2023). Ionic liquids based on 1-alkyl-3-methylimidazolium $[C_nmim]^+$ cations and bis(trifluoromethane-

sulfonyl)imide $[Tf_2N]^-$ anions are often employed alongside a range of extractant ligands, providing an organic phase that has a relatively low viscosity and can achieve high extraction percentages of REE (> 90 %) (Li and Angeli, 2016; Asadollahzadeh et al., 2020).

The high cost of IL has to date limited their industrial uptake. Small channel contactors offer a potential solution, where enhanced mass transfer rates and short residence times can reduce solvent demand and waste (Asrami et al., 2021). The large surface area-to-volume ratios, thin fluid films, and enhanced radial mixing increase mass transfer rates significantly compared to contactors such as mixer settlers (Zhang et al., 2014). Li and Angeli (2016), and Pheasey and Angeli (2023), applied IL in small channels for the extraction of Eu and Nd respectively, achieving extraction efficiencies of over 80 %. Van Stee et al. (2024), investigated multi-component mixtures using a multi-step approach to selectively separate Co and Ni in less than 20 mins.

The studies on the application of ionic liquids for intensified extractions of multi-component metal mixtures in small channels are limited, with no known works considering the selective separation of Nd from multi-component feeds. This work investigates the intensified selective extraction of Nd in small channels from Nd/Fe acidic solutions into the $[C_4mim][Tf_2N]$ - TOPO solvent phase. In a previous study, Rho et al. (2020), utilizing TOPO in kerosene for REE extraction from magnet swarfs, achieved 100 % extraction of the Nd, which was accompanied, however, by 10–45 % co-extraction of Fe. The results here indicate that when the IL is used, the extraction of Nd is selective and no Fe is co-extracted.

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2. Materials and methods

Iron(III) nitrate nonahydrate $Fe(NO_3)_3 \cdot 9H_2O$ (purity 99.9%, molecular weight 404 g/mol), neodymium (III) nitrate hexahydrate $Nd(NO_3)_3 \cdot 6H_2O$ (purity 99.9%, molecular weight 435.38 g/mol), the ionic liquid 1-butyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide ($[C_4mim][Tf_2N]$) (98 % purity), and the extractant trioctylphosphine oxide (TOPO) (purity 99%, molecular weight 386.63 g/mol) were purchased from Sigma Aldrich. For all experiments 3 repeats were made and the error was calculated as the standard deviation of the sample.

Rout et al. (2013) found that excessive acidity can hinder the cation exchange mechanism of Nd extraction by $[C_4mim][Tf_2N]$, thus reducing extraction efficiency. A 0.001 M nitric acid solution was, therefore, selected as the aqueous phase with neodymium (III) nitrate hexahydrate and iron(III) nitrate nonahydrate dissolved in it, producing two aqueous phase solutions, a single-component (0.01 M Nd), and a multi-component feed (0.01 M Nd and 0.05 M Fe), at concentrations found in NdFeB magnets (Schulze and Buchert, 2016). The organic phase in both cases was 0.12 M TOPO in $[C_4mim][Tf_2N]$. As the IL is hygroscopic, it was pre-equilibrated for 45 min with an equal volume of deionized water and separated with a separation funnel prior to experiments. The 0.12 M concentration of TOPO in the IL was selected to maintain a Nd:TOPO ratio greater than 1:6, which is the minimum needed according to the extraction mechanism, as shown by Pheasey and Angeli (2023).

Equilibrium experiments were undertaken to determine the maximum extraction percentage. 10 ml of each phase were brought into contact in a magnetically stirred sealed vessel and were mixed for 16 hours (960 min). Aqueous phase samples were removed for analysis every 15, 30, 60, 240, and 960 min (an equal volume of the organic phase was discarded to maintain the volume ratio), diluted to 10 - 100 ppm, and analyzed for Nd and Fe concentration by MP-AES (Agilent 4210) at wavelengths of 814 and 358.119 nm respectively. These were compared to a previously created calibration curve of known Nd or Fe concentrations (0 - 100 ppm) made using a Nd (Fe) standard solution of 1000 mg/L Nd (Fe) in 70 % nitric acid purchased from Merck. The extraction percentage % was then calculated according to Eq. 1.

$$\text{Extraction percentage(\%)} = \frac{M_{\text{initial}} - M_{\text{final}}}{M_{\text{initial}}} \times 100 \quad (1)$$

where M_{initial} is the initial aqueous phase metal concentration (Nd or Fe) and M_{final} is the final aqueous phase metal concentration (Nd or Fe).

Small channel experiments were performed in 0.5 mm circular PTFE tubing with a T-junction inlet. The aqueous phase entered perpendicular to the main channel while the organic phase was introduced tangentially as shown in Fig. 1. Experiments were undertaken in the plug flow regime, where aqueous drops form in the IL continuous phase with a diameter greater than the channel diameter (Fig. 1). Experiments were carried out at equal phase flow rates and a mixture velocity of 0.02 m/s giving plug length equal to 0.85 mm and an interfacial area equal to $4500 \text{ m}^2/\text{m}^3$ (Pheasey and Angeli, 2023). To obtain different residence times (7.5 to 37.5 s), the channel length was varied from 15 to 75 cm. At the end of the channel the mixture entered a membrane separator

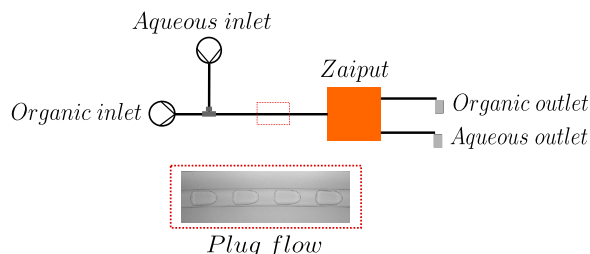


Fig. 1. Schematic of the small channel setup with in line membrane separation.

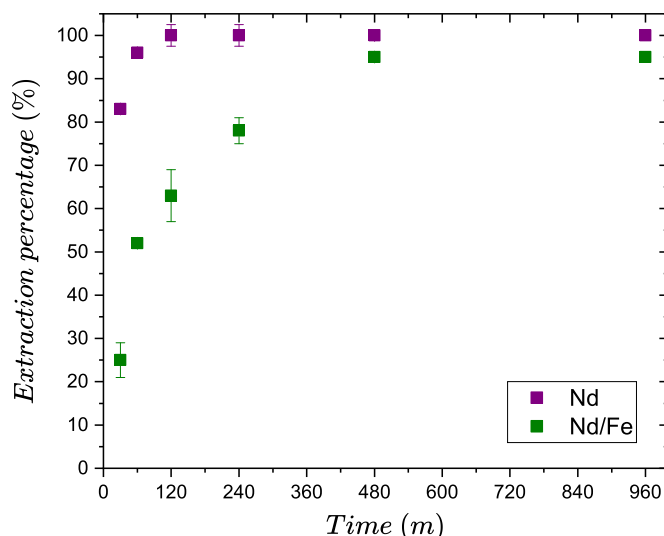


Fig. 2. Extraction percentage of Nd (calculated using Eq. 1) over time from an initial aqueous phase of Nd and Nd/Fe. Initial aqueous phase loadings 0.01 M Nd and 0.01 M Nd + 0.05 M Fe.

(Zaiput) fitted with a hydrophilic PTFE IL-2000-S10 membrane. The aqueous phase outlet was collected, diluted, and analyzed by MP-AES.

The interfacial tension of the fluids was 0.0118 N/m, measured with the pendant drop method using a DSA100E drop shape analyzer (KRÜSS Scientific) (accuracy 0.3 mN/M). The viscosity of the IL before and after it was saturated with water was 0.052 and 0.041 Pa.s respectively (Rheometer DV-111 Ultra (Brookfield) (+/- 1 %)).

3. Results and discussion

3.1. Equilibrium studies

The extraction of Nd from a single (Nd) and a multi-component (Nd/Fe) aqueous feed is shown in Fig. 2. Both aqueous feeds reach an equilibrium extraction of over 95 % (evidenced by the plateau in extraction) at 120 min for the single-component feed, and at 480 min for the multi-component feed. According to the Maxwell Stefan equation for diffusion (Bird et al., 2007) in multi-component systems the presence of additional solutes presents cross diffusional effects altering the diffusion coefficient and mass transfer. The presence of Fe^{3+} introduces ion-ion interactions that reduce the effective diffusivity of Nd^{3+} . In addition, the presence of Fe in the multi-component feed increases the ion strength of the solution reducing the diffusivity of Nd (Debye-Hückel Theory) (Atkins and De Paula, 2006). These effects reduce the extraction rate seen in Fig. 2.

For the multi-component feed, the concentration of Fe in the aqueous phase was also measured at each time interval and was found to remain constant. The primary complex of Nd in the aqueous phase, $[Nd(H_2O)_9]^{3+}$, is more weakly bound than the primary complex of Fe in $[Fe(H_2O)_6]^{3+}$, because of the larger ionic radius and lower charge density of Nd^{3+} compared to Fe^{3+} . This lowers the enthalpy of hydration of Nd^{3+} (3500 kJ/mol) compared to Fe^{3+} (4300 kJ/mol) (Marcus, 1987), which favours the dissociation of $[Nd(H_2O)_9]^{3+}$ over that of $[Fe(H_2O)_6]^{3+}$, resulting in the selective extraction of the Nd ions. Even when the Nd extraction reaches 100 % after 480 min, the Fe extraction remains at 0 %. It was found that between 0 and 960 min the aqueous phase changes colour from yellow to orange to brown (Fig. 3), suggesting a transition from $[Fe(H_2O)_6]^{3+}$ to hydroxide complexes according to Eq. 2. The iron ion in these complexes bonds strongly with hydroxide in Fe-O-Fe bonds (Cotton and Wilkinson, 1972) increasing the affinity of the Fe for the aqueous phase and preventing its extraction.

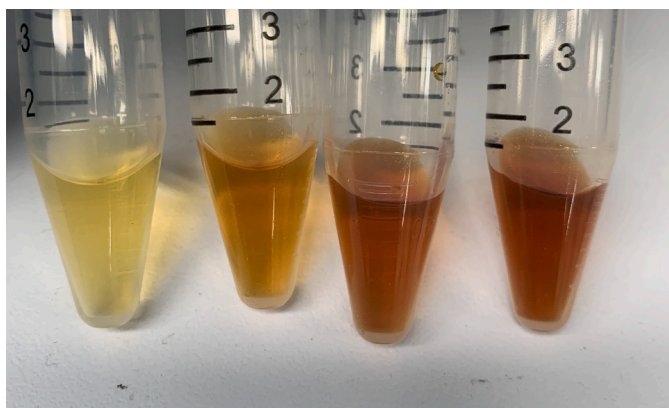
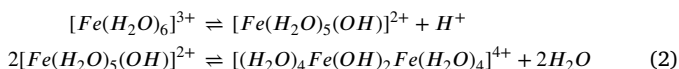


Fig. 3. Aqueous phase solution sample at 30, 60, 240, and 960 min.



3.2. Small channel extraction

Continuous flow extractions were investigated in small channels with both the single (0.01 M Nd) and the multi-component (0.01 M Nd and 0.05 M Fe) aqueous phase feeds and the 0.12 M TOPO/IL organic phase. Extraction efficiency (%), which indicates the extraction achieved relative to the maximum equilibrium extraction possible, and the overall volumetric mass transfer coefficient, $k_L\alpha$, were calculated as follows:

$$\text{Extraction efficiency (\%)} = \frac{C_{aq,in} - C_{aq,out}}{C_{aq,in} - C_{aq,eq}} \times 100 \quad (3)$$

$$K_L\alpha = \frac{1}{\tau} \ln \left(\frac{C_{aq,eq} - C_{aq,in}}{C_{aq,eq} - C_{aq,out}} \right) \quad (4)$$

where $C_{aq,in}$ is the concentration of the metal in the aqueous phase at the inlet, $C_{aq,out}$ is the concentration of the metal in the aqueous phase at the outlet, $C_{aq,eq}$ is the concentration of the metal in the aqueous phase at equilibrium from the batch studies, and τ is the residence time.

At low residence times (7.5 and 15 s), the extraction efficiencies are similar for both feeds at about 55 % (Fig. 4). In these cases, the mixing of the two phases at the channel inlet dominates the process. At the channel inlet, steep concentration gradients between the two phases combined with intense mixing (Chinaud et al., 2015) increase interface renewal and result in high mass transfer rates; these seem to overcome the limitations imposed by the presence of Fe in the mixture as seen in Fig. 2. With further increase of the residence time, the extraction efficiency for both aqueous feeds increases; the extended contact time between the phases allows greater interfacial mass transfer. This increase occurs at a reduced rate for the multi-component feed, resulting in a 12% less extraction efficiency at 37.5 s. The relatively high extraction efficiency of Nd (~ 55% of equilibrium) at the channel inlet increases the proportion of Fe in the aqueous phase downstream, which affects the Nd diffusivity compared to the single-component feed, thus reducing Nd extraction.

The volumetric mass transfer coefficients ($k_L\alpha$) for both initial aqueous feeds are presented in Fig. 5. At short residence times (7.5 s), both single and multi-component systems exhibit $k_L\alpha$ values of approximately 0.11 s^{-1} , attributed to the intense mixing conditions at the channel inlet. In comparison, the mass transfer coefficients for metal separations in stirred vessels are at least an order of magnitude lower, $0.053 - 0.013 \text{ s}^{-1}$ (Abbasi et al., 2018) and 0.0087 s^{-1} (Zhang et al., 2017). As the residence time increases, $k_L\alpha$ decreases for both systems and reaches a plateau, at 0.04 s^{-1} . At longer residence times (increased channel lengths) the effect of the inlet is small and plug flow is well established, resulting in almost constant mass transfer coefficients.

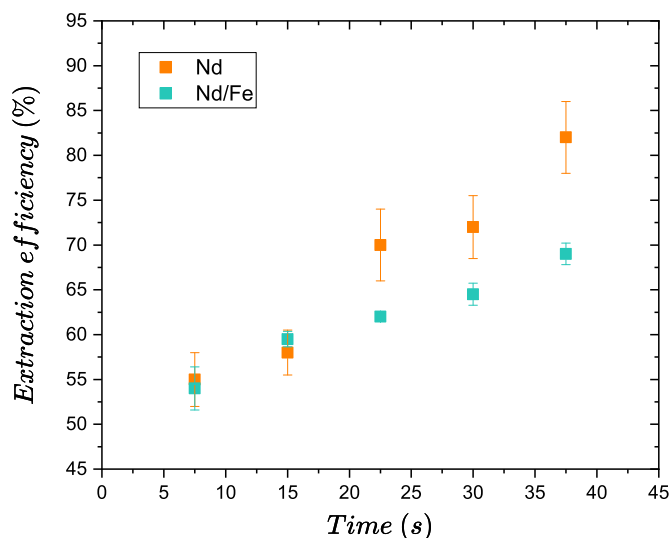


Fig. 4. Effect on extraction efficiency of increasing residence time, 7.5 - 37.5 s (increased channel length 15 - 75 cm) at a constant mixture velocity of 0.02m/s on extraction efficiency of Nd from a Nd and a Nd/Fe aqueous feed.

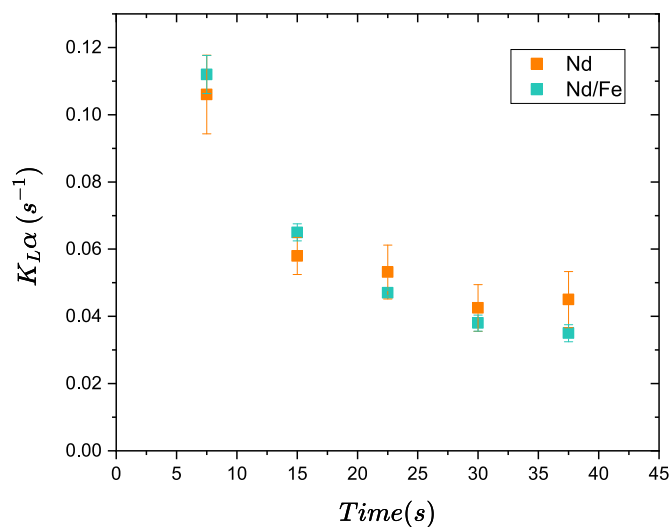


Fig. 5. Effect on mass transfer coefficient of increasing residence time, 7.5 - 37.5 s (increased channel length 15 - 75 cm) at a constant mixture velocity of 0.02m/s on $k_L\alpha$ of Nd from a Nd and a Nd/Fe aqueous feed.

4. Conclusions

This short communication presents new results on the selective extraction of Nd from a Nd/Fe aqueous feed into a $[C_4mim][Tf_2N]$ + TOPO solvent phase, at concentrations relevant to NdFeB magnet recycling.

Equilibrium studies showed significant differences in the Nd extraction between the Nd and Nd/Fe feeds, requiring 120 and 480 min respectively to reach maximum extraction. It was found that Fe did not extract to the ionic liquid phase even after 960 min. By applying an IL solvent phase with TOPO, 100 % selective extraction of the Nd was achieved. Intensified extractions were possible in small channels. High extraction efficiencies were obtained for both aqueous feeds, reaching 80 % and 65 %, respectively in very short residence times of ~ 37 s. The mass transfer coefficients ranged from $0.1 - 0.04 \text{ s}^{-1}$ at 7.5 - 37.5 s respectively.

These results demonstrate the applicability of the IL combined with the TOPO extractant for the 100 % selective Nd extraction from a Nd/Fe

feed and the intensification of the process in small channels. Future studies will utilize a magnet swarf and apply stripping of the elements for recovery and solvent recycle.

CRediT authorship contribution statement

Charlotte Pheasey: Conceptualization, Data curation, Formal analysis, Visualization, Writing – original draft; **Panagiota Angeli:** Writing – review & editing, Supervision, Resources, Project administration.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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